

Custom STM32 Dev Board

4-layer PCB with USB-C PD negotiation, CAN bus interface, and isolated analog front-end for industrial sensor logging.

KiCad	STM32	C/C++	CAN Bus
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REVISION	STATUS	STACK-UP	MCU	ISOLATION	UPDATED
Rev C	Complete	4 layer	STM32G474	3 kV rms	Sep 2026

1. Overview

A four-layer development board for industrial sensor logging. It draws power over USB-C with full Power Delivery negotiation, talks to equipment over an isolated CAN FD interface, and samples analogue sensors through a galvanically isolated 24-bit front end. The intent was a board that can sit on a machine rather than only on a bench.

Isolation is the organising constraint. The analogue front end and the CAN transceiver each sit on their own isolated domain, fed by isolated DC-DC converters, so a ground potential difference between the sensor, the bus, and the host laptop cannot inject current through the measurement path or damage the USB host. The board is at Rev C, which is the revision that went into service; no further spins are planned.

2. Block diagram

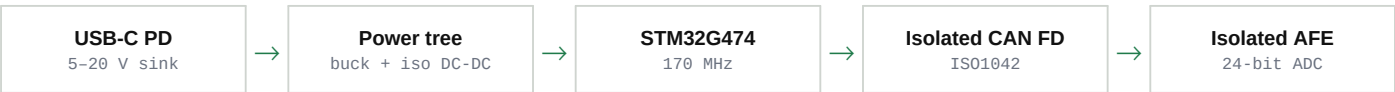


Figure 1 — Two isolated domains hang off the digital core, each with its own supply and digital isolator; no signal crosses a barrier unisolated.

Power delivery negotiation is handled by a standalone PD sink controller rather than in firmware, so the board reaches a stable rail before the MCU is released from reset. It requests 15 V and falls back through 9 V to 5 V; at 5 V the analogue domain stays enabled but the high-side sensor excitation supply is disabled and reported over USB.

3. Board sections

Power input	USB-C receptacle, PD sink controller, input TVS and reverse protection, 20 V rating.
Power tree	Synchronous buck to 5 V and 3.3 V, plus two isolated DC-DC modules for the barriers.
Digital core	STM32G474 at 170 MHz, 8 MB QSPI flash, microSD on SDMMC, SWD and USB CDC.
CAN interface	Isolated CAN FD transceiver, split termination with a jumper, bus fault protection.
Analogue front end	Four differential channels, ± 10 V and 4–20 mA, 24-bit delta-sigma ADC.
Protection	ESD arrays on all external connectors; PTC on the excitation supply output.

4. Electrical specification

PARAMETER	VALUE	NOTES
Input	USB-C PD, 5–20 V, 45 W max	Fixed PDO request 15 V, graceful fallback.
Analogue inputs	4 $\times$ differential, ± 10 V or 4–20 mA	Per-channel jumper selects mode.
Sample rate	32 kSPS max, 1 kSPS typical	Simultaneous sampling across four channels.
CAN	CAN FD to 5 Mbit/s data phase	Isolated; classic CAN supported for legacy buses.

5. Firmware bring-up

```
/* Rev C pin assignment, verified against the fabricated board. */
#define PD_ALERT_PIN    GPIOB, 2      /* PD sink controller interrupt */
#define CAN_TX_PIN      GPIOB, 9      /* FDCAN1 through ISO1042 */
#define CAN_RX_PIN      GPIOB, 8
#define ADC_DRDY_PIN    GPIOC, 6      /* ADS131M04 data ready, isolated */
#define ADC_CS_PIN      GPIOC, 7      /* SPI2, 8 MHz across the barrier */
#define EXC_EN_PIN      GPIOA, 15     /* 24 V sensor excitation enable */

void board_init(void) {
    clock_config_170mhz_pll();
    pd_sink_wait_contract(&pd, PD_TIMEOUT_MS); /* rails settled first */
    if (pd.voltage_mv < 9000) exc_supply_disable();

    fdcan_init(FDCAN_NOMINAL_500K, FDCAN_DATA_2M);
    adc_init(ADS131M04_OSR_1024);             /* 1 kSPS, 4 channels */
    adc_start_continuous(on_drdy);             /* DMA into ring buffer */
}
```

Listing 1 — Excitation is gated on the negotiated voltage, so a 5 V-only host cannot brown out the analogue domain.

6. Layout notes

- **Stack-up.** Signal, ground, power, signal on 1.6 mm FR-4. The uninterrupted ground plane directly under the digital core is what brought radiated emissions within margin on Rev C after Rev B split it for a power pour.
- **Isolation barriers.** 4.5 mm clearance under each isolator with no copper, no silkscreen, and no inner-layer crossing, supporting the 3 kV rms rating of the chosen parts.
- **Analogue routing.** Differential pairs routed as a tight pair over a dedicated analogue ground pour, with the ADC reference bypassed at the pin and the switching regulator placed at the opposite corner of the board.

7. Measured performance

MEASUREMENT	RESULT	CONDITIONS
Noise floor	2.1 μ V rms	Inputs shorted, 1 kSPS, $\pm$ 10 V range.
Effective resolution	20.4 bits	Derived from noise floor, same conditions.
PD negotiation	310 ms to stable rail	Cold plug, 15 V contract.
Idle current	84 mA at 15 V	MCU running, CAN idle, ADC converting.

8. Manufacturing and test

- Assembled in small runs from a single KiCad project with the BOM and pick-and-place exported directly; all parts are single-source only where no alternate exists, noted in the BOM.
- Every board runs a bring-up script over SWD that checks rail voltages, walks the PD voltage ladder, loops back CAN through the on-board termination, and records ADC noise with inputs shorted. Results are stored against the board serial.
- Test points are exposed for each rail and both isolated grounds, which is what made Rev B's isolated DC-DC ripple problem findable without cutting traces.

9. Maintenance status

AREA	STATE	COMMITMENT
Hardware revisions	Closed	Rev C is final; no further board spins planned.
Firmware	Limited	Bug fixes only for the reference application.
Documentation	Available	Schematic, gerbers, and BOM published with the project.